COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	09/17/2026	**DATE SUBMITTED:**	09/17/2026
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HANDS ON ACTIVITY #9.1 **Tree ADT**

1. PROCEDURE OUTPUT

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Step 1: Implement a binary search tree that will take the following values: 2, 3, 9, 18, 0, 1, 4, 5.

main.cpp

```cpp
#include <iostream>
#include "BST.h"

int main(){

    BinarySearchTree<int>* tree = new BinarySearchTree(2);
    tree->insert(3);
    tree->insert(9);
    tree->insert(18);
    tree->insert(0);
    tree->insert(1);
    tree->insert(4);
    tree->insert(5);

    std::cout << "Pre Order Output: ";
    tree->preOrder();

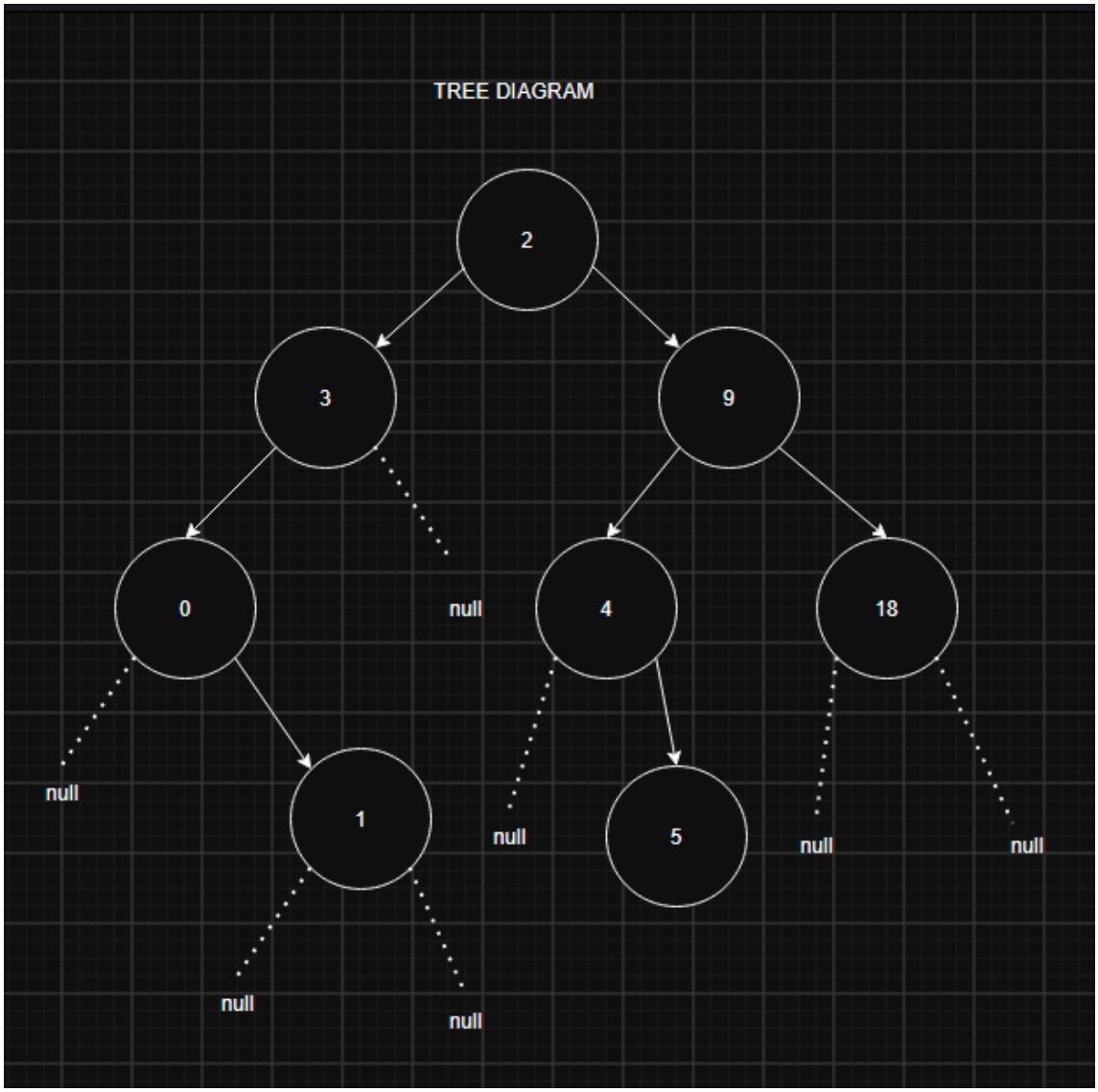
    std::cout << "\nIn Order Output: ";
    tree->inOrder();

    std::cout << "\nPost Order Output: ";
    tree->postOrder();

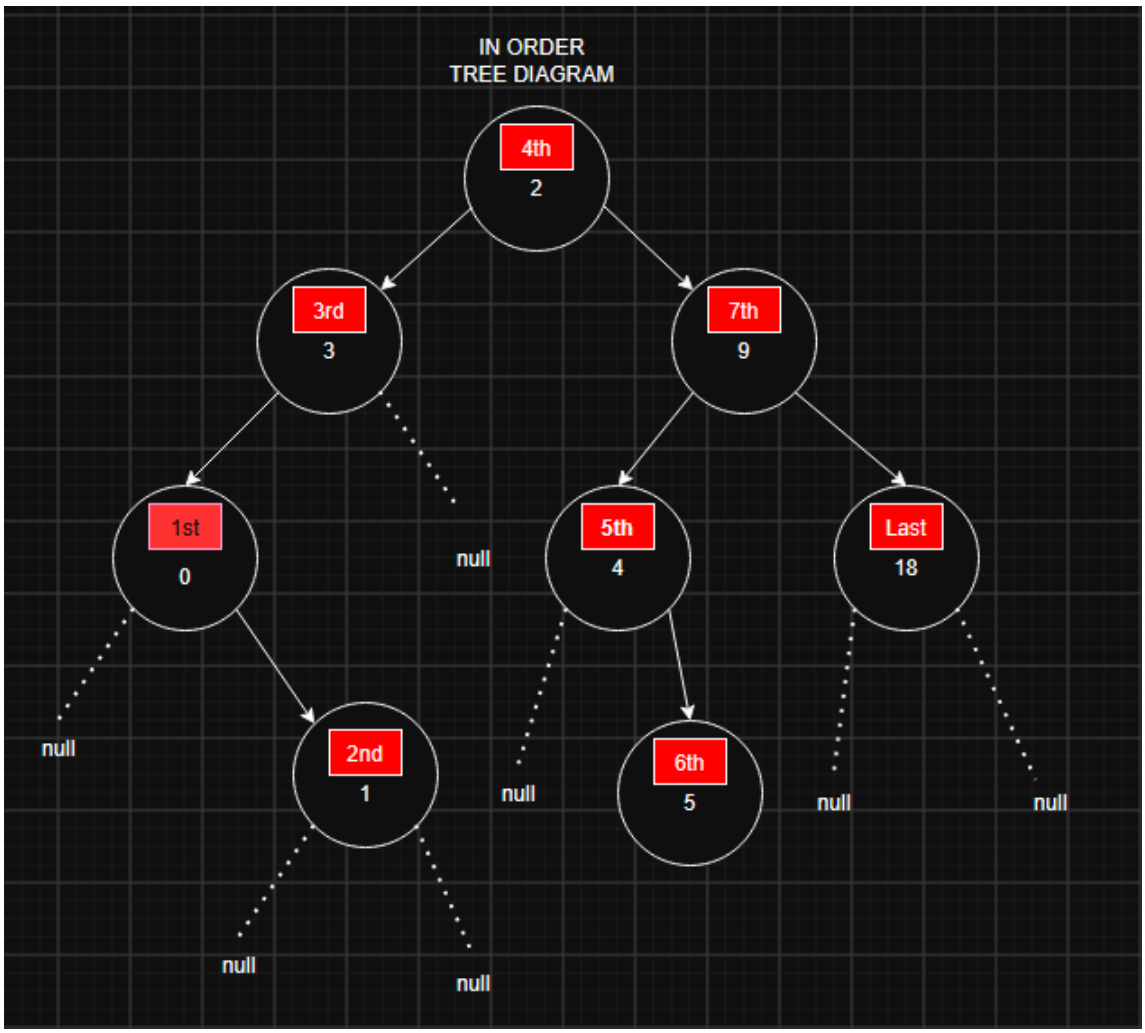
    return 0;
}
```

Step 2: Create a diagram to show the tree after all values have been inserted. Then, with the use of visual aids (like arrows and numbers) indicate the traversal order for in-order, pre-order and post-order traversal on the diagram.

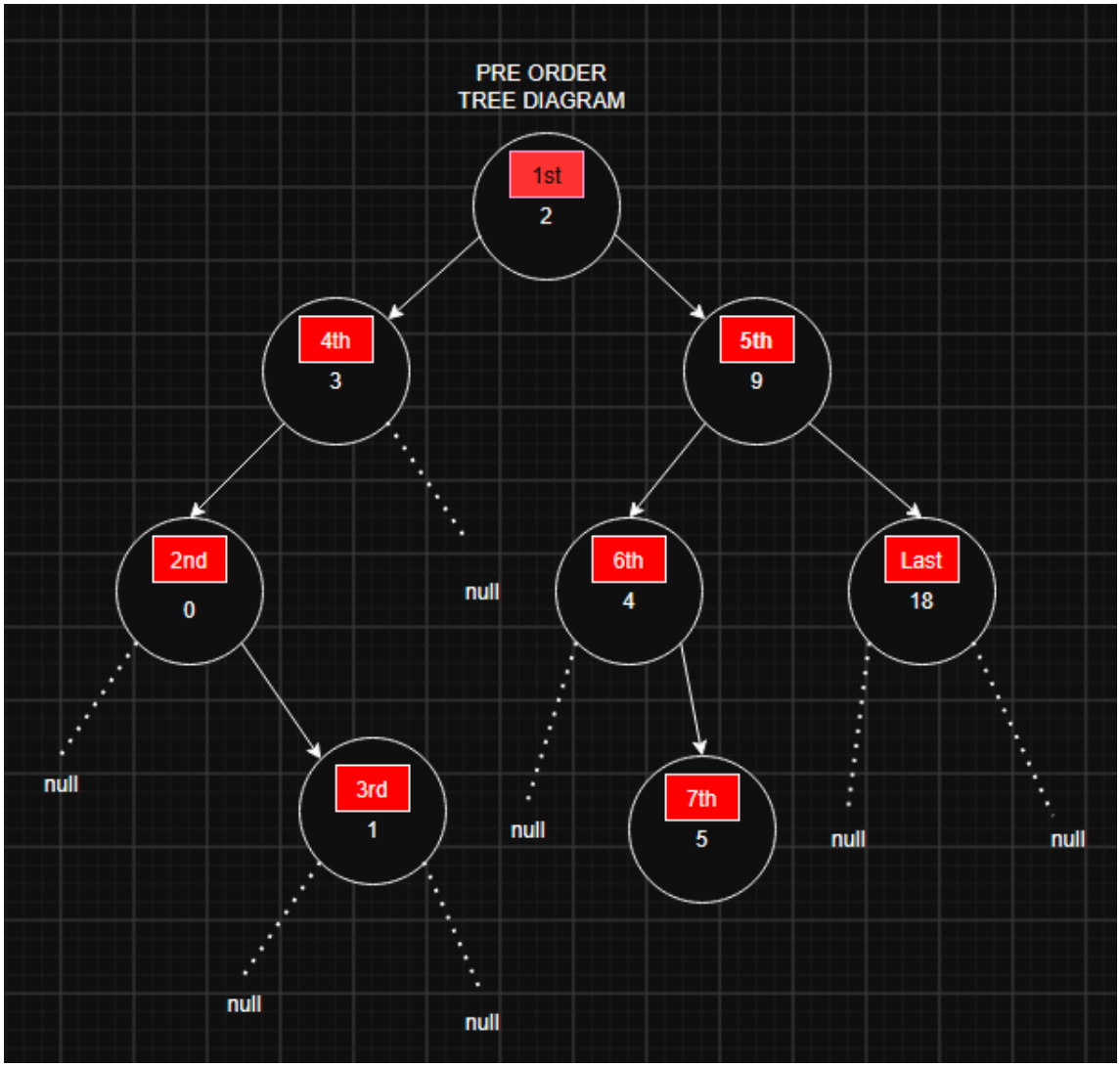
Tree diagram:



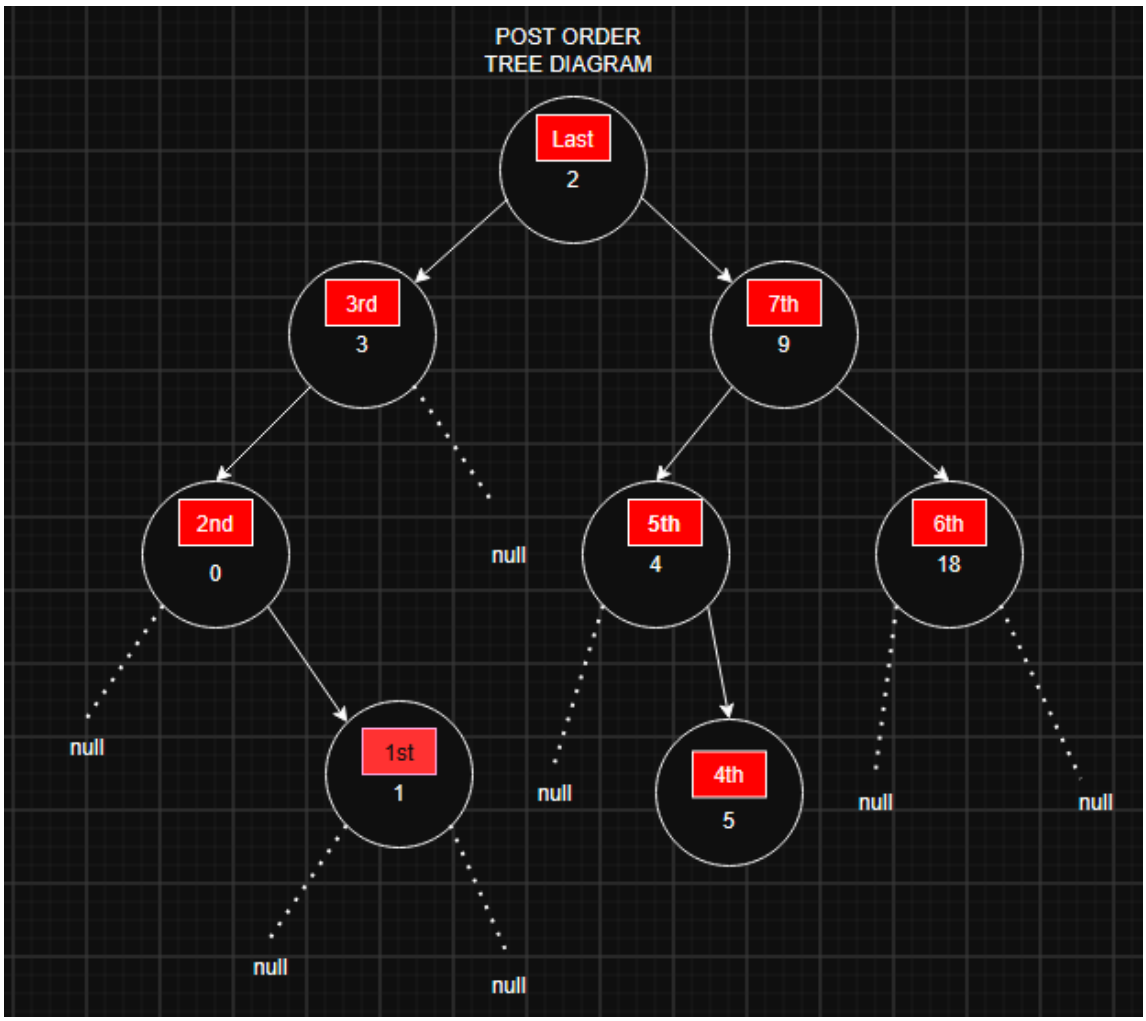
In Order diagram:



Pre Order diagram:



Post Order diagram:



Step 3: Compare the different traversal methods. In-order traversal was performed with what function?

In-Order:

```
66   //in order def
67   template <typename T>
68   void BinarySearchTree<T>::inOrder(){
69       //traversal
70       if (left != nullptr){
71           left->inOrder();
72       }
73       //print root
74       std::cout << key << " ";
75       //traversal
76       if (right != nullptr){
77           right->inOrder();
78       }
79   }
```

Output:

```
In Order Output: 0 1 2 3 4 5 9 18
```

Pre-Order:

```
//pre order def
template <typename T>
void BinarySearchTree<T>::preOrder(){
    //print root
    std::cout << key << " ";
    //traversal
    if (left != nullptr){
        left->preOrder();
    }
    if (right != nullptr){
        right->preOrder();
    }
}
```

Output:

```
Pre Order Output: 2 0 1 3 9 4 5 18
```

Post Order:

```cpp
//post order def
template <typename T>
void BinarySearchTree<T>::postOrder(){
    //traversal
    if (left != nullptr){
        left->postOrder();
    }
    if (right != nullptr){
        right->postOrder();
    }
    //print root
    std::cout << key << " ";
}
```

Output:

```
Post Order Output: 1 0 5 4 18 9 3 2
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

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